Boston Heart & Vascular Institute

110 Francis St, Suite 4B, Boston, MA 02215
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Patient: Frank J. Donnelly
DOB: 08/28/1950
MRN: BHVI-0044028

Date: 01/26/2026
Provider: Thomas Greer, MD, FACC
Specialty: Cardiovascular Disease / Interventional Cardiology

CHIEF COMPLAINT

Referred by PCP after transient episode of right arm weakness and slurred speech.

HISTORY OF PRESENT ILLNESS

Mr. Donnelly is a 75-year-old male with a history of hypertension, hyperlipidemia, type 2 diabetes, and known coronary artery disease (stent to LAD, 2019) referred for cardiology evaluation following an episode of sudden right arm weakness and dysarthria that lasted approximately 25 minutes and then fully resolved. Episode occurred on January 18, 2026, while he was reading the newspaper. His wife noted slurred speech and saw him drop his coffee cup. He had no headache, no vision changes, and no loss of consciousness. He did not call 911; his PCP saw him the next day.

His PCP obtained an MRI brain (results reviewed below) and referred to both Cardiology and Neurology. He is on aspirin 81 mg, clopidogrel 75 mg (from his coronary stent), atorvastatin 80 mg, lisinopril 20 mg, metoprolol succinate 50 mg, and metformin 1000 mg BID. He has had no prior TIA or stroke. He was a smoker for 30 years, quit 10 years ago.

PHYSICAL EXAM

Vitals: BP 152/90, HR 68, regular. Wt 198 lbs.
Cardiovascular: RRR, no murmurs or gallops. No JVD. Trace bilateral ankle edema.
Carotids: Right carotid bruit audible.
Neuro (today): No focal deficits. Normal cranial nerves, strength, speech.

RESULTS REVIEWED

MRI brain with DWI (01/20/2026): Small area of restricted diffusion in the left pons — see radiology report. Consistent with small acute infarction.
ECG today: Normal sinus rhythm, HR 68. No ST changes. No LVH. QTc 418 ms.

ASSESSMENT AND PLAN

1. TIA / minor ischemic stroke, left pontine (G45.9 / I63.9)
Despite symptom resolution, MRI evidence of acute pontine infarct makes this a minor stroke rather than a pure TIA. The right carotid bruit raises concern for significant ipsilateral (left-sided) stenosis given contralateral symptom pattern — or alternatively the bruit may represent the right side while pathology is on the left.

Carotid duplex ultrasound ordered (see results). Echocardiogram ordered to evaluate for cardioembolic source (LV thrombus, structural disease, PFO). Hypercoagulability panel ordered given age and cryptogenic component if stenosis is insufficient to explain.

Not changing antithrombotic regimen today — he is already on dual antiplatelet therapy for his coronary stent. Coordinate with Neurology regarding definitive antithrombotic strategy. If PFO found, anticoagulation vs. device closure discussion will be needed.

2. Hypertension, inadequately controlled (I10)
BP 152/90 — target <130/80 post-stroke per ACC/AHA. Increase lisinopril to 40 mg. Add amlodipine 5 mg. Recheck in 2 weeks.

3. Neurology referral
Urgent neurology referral in place per PCP. I have called Dr. Hasan directly to discuss the case and share imaging results. Will co-manage.

Electronically signed by: Thomas Greer, MD, FACC
Signed 01/26/2026 14:40 EST | BHVI Cardiology EHR | cc: Neurology — Dr. Hasan

Massachusetts General Hospital — Department of Radiology
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RADIOLOGY REPORT

Patient: Frank J. Donnelly	**DOB:** 08/28/1950	**MRN:** BHVI-0044028
Exam Date: 01/20/2026 14:10	**Ordering MD:** Robert Klein, MD (PCP)	**Accession:** MGH-MR-20260120-8812

EXAM

MRI BRAIN WITH DIFFUSION-WEIGHTED IMAGING, WITHOUT AND WITH GADOLINIUM CONTRAST

CLINICAL INDICATION

75-year-old male with acute-onset right arm weakness and dysarthria lasting 25 minutes, now resolved. Rule out acute infarction.

TECHNIQUE

3T MRI brain: axial DWI/ADC, axial FLAIR, axial T2, axial T2*, axial and sagittal T1, post-contrast axial T1 MPRAGE. Gadavist 0.1 mmol/kg administered.

COMPARISON

No prior MRI available.

FINDINGS

Diffusion-weighted imaging: A 7 x 5 mm focus of restricted diffusion (bright on DWI, dark on ADC map) is identified in the left paramedian pons at the level of the mid-pons. This is consistent with acute to early subacute infarction (approximately 2-7 days old).

FLAIR: The pontine lesion demonstrates early T2/FLAIR signal change. No other areas of restricted diffusion. No additional acute lesions identified.

White matter: Moderate periventricular and subcortical white matter T2/FLAIR hyperintensities in a pattern consistent with chronic small vessel ischemic disease (Fazekas grade 2). This is an expected finding for age.

Contrast enhancement: No abnormal enhancement identified. The pontine lesion does not enhance (consistent with acute/early subacute ischemia rather than demyelination or neoplasm).

Major vessels: Flow voids of the basilar artery, bilateral ICAs, and MCAs appear grossly preserved on this non-MRA study. MRA is recommended for vascular assessment.

Posterior fossa: No cerebellar infarction. Brainstem otherwise unremarkable outside the described lesion. No mass or herniation.

Ventricles/Calvarium: Mild prominence of cortical sulci consistent with age-related volume loss. No hydrocephalus. No hemorrhage. No calvarial abnormality.

IMPRESSION

1. 7 x 5 mm acute to early subacute infarction, left paramedian pons. Clinically consistent with the reported episode of right arm weakness and dysarthria (corticospinal and corticobulbar tract involvement at the mid-pons level).
2. Moderate chronic small vessel ischemic changes (Fazekas grade 2) — background finding consistent with patient's vascular risk factor history.
3. No intracranial hemorrhage.
4. MRA of the head and neck is recommended to evaluate the posterior circulation vessels (basilar artery, vertebral arteries) and carotid arteries.

Electronically attested by: Patricia Walsh, MD — Neuroradiology
Attested 01/20/2026 17:50 EST | MGH Radiology PACS

Boston Heart & Vascular Institute — Vascular Lab

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CAROTID DUPLEX ULTRASOUND REPORT

Patient: Frank J. Donnelly	**DOB:** 08/28/1950	**MRN:** BHVI-0044028
Exam Date: 01/27/2026 09:30	**Ordering MD:** Thomas Greer, MD	**Accession:** BHVI-VL-20260127-0441

CLINICAL INDICATION

75-year-old male with acute left pontine infarct. Right carotid bruit on exam. Evaluate carotid stenosis.

TECHNIQUE

B-mode and color/spectral Doppler ultrasound of bilateral carotid arteries including common carotid, carotid bifurcation, proximal internal carotid (ICA), and external carotid arteries (ECA). Vertebral artery flow also assessed.

FINDINGS

RIGHT CAROTID SYSTEM:
Common carotid artery (CCA): Patent, normal flow. No plaque.
Carotid bifurcation: Heterogeneous calcified and soft plaque at the right carotid bulb extending into the proximal ICA. Plaque surface irregular.
Internal carotid artery (ICA):
  PSV: 248 cm/s; EDV: 112 cm/s
  ICA/CCA ratio: 4.2
  Estimated stenosis: 70-79% (moderate-severe by NASCET criteria)
External carotid artery: Patent, no significant stenosis.

LEFT CAROTID SYSTEM:
Common carotid artery: Patent, normal flow.
Carotid bifurcation: Mild intima-media thickening. Small calcified plaque at bifurcation.
Internal carotid artery (ICA):
  PSV: 98 cm/s; EDV: 34 cm/s
  ICA/CCA ratio: 1.4
  Estimated stenosis: <50% (mild)

VERTEBRAL ARTERIES:
Bilateral vertebral artery flow is antegrade and symmetric. No significant stenosis identified.

IMPRESSION

1. Right internal carotid artery stenosis 70-79% (NASCET moderate-severe) with heterogeneous, irregular plaque — high-risk plaque morphology.
2. Left ICA stenosis less than 50% — mild, not hemodynamically significant.
3. Normal vertebral artery flow bilaterally.

Note: The acute infarction was in the left pons (posterior circulation territory). The right ICA stenosis does not directly explain a left pontine infarct via the carotid system. CTA or MRA of the posterior circulation (vertebrobasilar system) is recommended to evaluate for vertebral artery or basilar artery disease as the more likely culprit vessel. Right ICA stenosis warrants vascular surgery consultation regardless for secondary prevention.

Electronically attested by: Kevin Moore, RVT — Registered Vascular Technologist
Interpreted by: Thomas Greer, MD, FACC | Attested 01/27/2026 12:15 EST

Boston Heart & Vascular Institute — Echocardiography Lab

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ECHOCARDIOGRAPHY REPORT — TRANSTHORACIC + BUBBLE STUDY

Patient: Frank J. Donnelly	**DOB:** 08/28/1950	**MRN:** BHVI-0044028
Exam Date: 01/27/2026 11:00	**Ordering MD:** Thomas Greer, MD	**Accession:** BHVI-ECHO-20260127-0214

CLINICAL INDICATION

Acute left pontine infarct. Evaluate for cardioembolic source. Bubble study for PFO.

TECHNIQUE

Complete 2D transthoracic echocardiogram with Doppler. Agitated saline contrast (bubble study) performed at rest and with Valsalva maneuver.

FINDINGS

Left ventricle: Normal size. EF estimated 55-60% (normal). No wall motion abnormalities. No LV thrombus. Normal diastolic function (Grade I impairment).
Right ventricle: Normal size and systolic function.
Atria: Left atrium mildly dilated (LA volume index 32 mL/m2; normal <34). Right atrium normal size. No intracardiac thrombus identified. No spontaneous echo contrast.
Valves: Mild mitral annular calcification. Mild mitral regurgitation (1+). Aortic valve mildly thickened and calcified; aortic stenosis mild (peak gradient 18 mmHg, mean 9 mmHg). Tricuspid valve normal. No pulmonic stenosis.
Pericardium: No pericardial effusion.
Aortic root: Mildly dilated at 3.9 cm.

BUBBLE STUDY (Agitated Saline Contrast):
At rest: No right-to-left shunting observed.
With Valsalva: POSITIVE — appearance of >20 microbubbles in the left atrium within 3 cardiac cycles of right atrial opacification. This is consistent with a patent foramen ovale (PFO) with right-to-left shunting on Valsalva. Bubble passage quantity suggests a moderate-sized PFO.

IMPRESSION

1. Normal left ventricular size and systolic function (EF 55-60%). No LV thrombus.
2. Mild LA enlargement. No intracardiac mass or thrombus.
3. Mild valvular disease as described (mitral regurgitation, early aortic stenosis) — not considered embolic source.
4. PATENT FORAMEN OVALE (PFO) with moderate right-to-left shunting demonstrated on Valsalva — potentially relevant embolic source, particularly in context of acute ischemic stroke. Correlation with clinical stroke characteristics and discussion of closure vs. anticoagulation vs. continued antiplatelet therapy is recommended in a multidisciplinary setting.

Brigham Specialized Coagulation Laboratory

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HYPERCOAGULABILITY / THROMBOPHILIA PANEL

Patient: Frank J. Donnelly	**DOB:** 08/28/1950	**MRN:** BHVI-0044028
Exam Date: 01/27/2026 08:00	**Ordering MD:** Thomas Greer, MD	**Accession:** BCL-20260127-5541

CLINICAL INDICATION: Acute ischemic stroke — evaluate for hereditary or acquired thrombophilia.

NOTE: Patient is on aspirin 81 mg and clopidogrel 75 mg at time of collection. Antiplatelet agents may affect some functional assays; results should be interpreted in clinical context. Patient is NOT on anticoagulation.

ANTIPHOSPHOLIPID ANTIBODY PANEL

Test	Result	Reference	Flag
Anticardiolipin IgG	8 GPL	<20 GPL (negative)	
Anticardiolipin IgM	6 MPL	<20 MPL (negative)	
Anti-beta2-glycoprotein I IgG	4 SGU	<20 SGU (negative)	
Anti-beta2-glycoprotein I IgM	3 SMU	<20 SMU (negative)	
Lupus Anticoagulant (dRVVT screen)	Negative	Negative	
Lupus Anticoagulant (Silica clot time)	Negative	Negative	

HEREDITARY THROMBOPHILIA

Test	Result	Reference	Flag
Factor V Leiden mutation (PCR)	HETEROZYGOUS (R506Q)	Negative / Wild-type	H
Prothrombin G20210A mutation (PCR)	Not detected	Negative	
MTHFR C677T	Heterozygous	—	
MTHFR A1298C	Not detected	—	
Protein C activity	88%	70-140%	
Protein S (free antigen)	74%	65-140%	
Antithrombin III activity	96%	80-120%	
Homocysteine (fasting)	18.4 µmol/L	<15 µmol/L	H

OTHER COAGULATION

Test	Result	Reference	Flag
PT / INR	12.1 sec / 1.0	11-14 sec / 0.8-1.2	
aPTT	30 sec	25-38 sec	
Fibrinogen	388 mg/dL	200-400 mg/dL	
D-dimer	0.62 mg/L FEU	<0.50 mg/L FEU	H

INTERPRETATION: Factor V Leiden heterozygous mutation detected. This is the most common inherited thrombophilia and confers approximately 3-7x increased risk of venous thromboembolism; arterial risk data are less robust but some studies suggest modest increased stroke risk. Mildly elevated homocysteine (18.4) is a modifiable risk factor — supplement with folate and B6/B12. MTHFR heterozygosity may contribute to homocysteinemia. No antiphospholipid antibodies detected. D-dimer mildly elevated — nonspecific in the context of recent ischemic event.

Electronically attested by: Anna Bergstrom, PhD — Hemostasis Laboratory
Director: Robert Flaumenhaft, MD | Reported 01/27/2026 15:30 EST